MODULE 1 · LECTURE 11

Computer Models in Agriculture: Statistical Models, Weather Analysis and Crop Simulation Models

What a model is and how it is built and tested; statistical models that link yield to weather and inputs; analysing and forecasting weather for farming; and how crop simulation models such as DSSAT, APSIM, AquaCrop and InfoCrop grow a virtual crop day by day.

Learning objectives

After this lecture you should be able to:

1. Define a model and classify agricultural models.
2. Describe the steps in building a model, including calibration, validation and sensitivity analysis.
3. Fit and interpret a simple regression model linking yield to weather.
4. Analyse weather data: rainfall probability, dependable rainfall and growing degree days.
5. Explain IMD forecast ranges and agromet advisory services.
6. Describe the structure, inputs and outputs of crop simulation models and compare DSSAT, APSIM, AquaCrop and InfoCrop.
7. Evaluate model predictions using MBE, RMSE and the index of agreement.

Scope: Lecture 5 introduced spreadsheet formulas and trendlines, and Lecture 8 programming. This lecture shows how such tools become models of crops and weather. Crop water and nutrient tools follow in Lecture 13, decision support systems in Lecture 16, and data-driven AI models in Lectures 19–28.

1. What is a model?

A **model** is a simplified representation of a real system that helps us understand it or predict its behaviour. A **computer model** expresses the system as equations and rules in a program, so that "what if?" questions can be answered quickly — what if sowing is delayed by two weeks, if rainfall is 20% below normal, or if nitrogen is split into three doses? Agricultural systems modelling has developed over more than six decades, drawing on agronomy, soil science, meteorology, economics and computing (Jones et al., 2017).

Types of models

Classification	Types	Example
How the model is built	**Empirical (statistical)** — fits relationships to observed data. **Mechanistic (process-based)** — represents the underlying physical and biological processes	Yield regressed on rainfall / a crop model simulating photosynthesis and water uptake
Role of chance	**Deterministic** — the same inputs always give the same output. **Stochastic** — includes random variation and gives probabilities	A crop model run for one season / a weather generator producing many possible seasons
Role of time	**Static** — no change over time. **Dynamic** — state variables change step by step	A fertiliser-response equation / a daily crop-growth model
Purpose	**Descriptive** — describes behaviour. **Explanatory** — explains why. **Optimisation** — finds the best option	Trend line / crop model / linear programming for farm planning
Scale	Plant, field, farm, region, country	Field-level irrigation model / district yield forecast

2. Building and testing a model

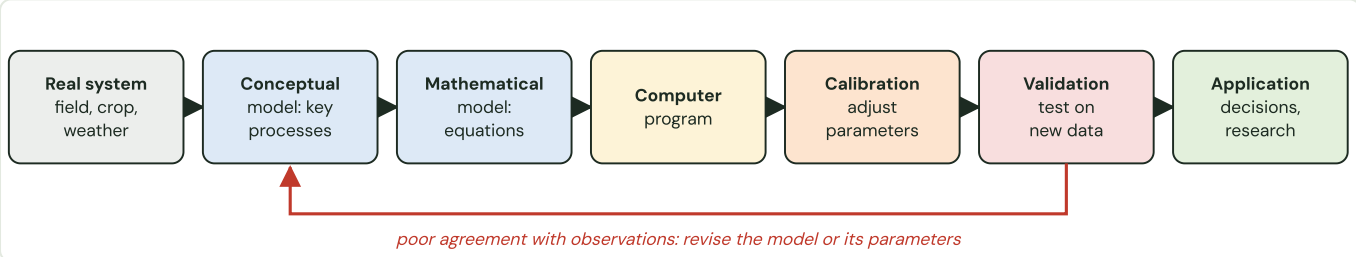


Figure 1. Steps in developing a computer model. Original diagram.

- **Conceptual model** — decide which processes matter and how they are linked.
- **Mathematical model** — write equations for each process.
- **Computer program** — code the equations (Lecture 8), usually with a daily or hourly time step.
- **Calibration** — adjust uncertain parameters, such as a cultivar's days to flowering, so that the model matches one set of observations.
- **Validation** — test the calibrated model against **independent** data that were not used for calibration.
- **Sensitivity analysis** — change one input or parameter at a time to see which ones affect the results most; these need the most accurate data.

Agreement between predicted (P) and observed (O) values is measured with simple statistics:

$$\mathrm{MBE} = \Sigma(P_i - O_i) \,/\, n \qquad \mathrm{RMSE} = \sqrt{[\,\Sigma(P_i - O_i)^2 \,/\, n\,]} \qquad \mathrm{nRMSE} = \mathrm{RMSE} \,/\, \bar{O} \times 100\%$$

$$d = 1 - \Sigma(P_i - O_i)^2 \,/\, \Sigma(\,|P_i - \bar{O}| + |O_i - \bar{O}|\,)^2$$

The **mean bias error (MBE)** shows whether the model over-predicts (positive) or under-predicts (negative) on average; the **root mean square error (RMSE)** shows the typical size of the error in the same units as the data; and Willmott's **index of agreement (d)** ranges from 0 (no agreement) to 1 (perfect agreement) (Willmott, 1981).

Worked example 1 — evaluating a crop model (illustrative values)
Observed wheat yields at five sites (t/ha): 4.2, 3.8, 5.1, 4.6, 3.5; simulated: 4.0, 4.1, 4.8, 4.9, 3.3. Mean observed $\bar{O}$ = 4.24 t/ha.
Differences P − O: −0.2, +0.3, −0.3, +0.3, −0.2 → MBE = −0.1 / 5 = −0.02 t/ha (almost no bias).
Squared differences: 0.04, 0.09, 0.09, 0.09, 0.04 → sum 0.35 → RMSE = √(0.35 / 5) = 0.26 t/ha; nRMSE = 0.26 / 4.24 = 6.2%.
Denominator of d: $\Sigma(|P - \bar{O}| + |O - \bar{O}|)^2$ = 6.294 → d = 1 − 0.35 / 6.294 = 0.94.
The model reproduces these yields closely; a small RMSE and a d close to 1 indicate good agreement.

3. Statistical models

Statistical models describe relationships in data without simulating the processes behind them. They need less information than process models and are widely used in field experiments, forecasting and planning (Gomez & Gomez, 1984).

Model or method	Used for	Agricultural example
Correlation	Strength and direction of a linear relationship (r from −1 to +1)	Rainfall and yield; soil organic carbon and nitrogen
Simple and multiple linear regression	Predicting one variable from one or more others	Yield from rainfall, temperature and fertiliser
Polynomial (e.g. quadratic) regression	Curved responses with an optimum	Yield response to nitrogen dose
Analysis of variance (ANOVA)	Comparing treatments in designed experiments	Comparing varieties in a randomised block design
Time-series models	Trends, seasonality and forecasts from past values (moving averages, ARIMA)	Monthly mandi prices; long-term rainfall trend
Probability distributions	Chance of an event of a given size	Dependable rainfall; flood and drought frequency
Machine learning	Complex, non-linear patterns in large data	Yield prediction from satellite and weather data (Lectures 19–28)

Simple linear regression

$$\hat{y} = a + b \cdot x \qquad b = \Sigma(x - \bar{x})(y - \bar{y}) \,/\, \Sigma(x - \bar{x})^2 \qquad a = \bar{y} - b \cdot \bar{x}$$

$$r = \Sigma(x - \bar{x})(y - \bar{y}) \,/\, \sqrt{[\,\Sigma(x - \bar{x})^2 \cdot \Sigma(y - \bar{y})^2\,]} \qquad R^2 = r^2$$

Here **b** is the slope (change in yield per unit of x), **a** the intercept, **r** the correlation coefficient, and **$R^2 = r^2$** the fraction of the variation in y explained by x.

Worked example 2 — a yield–rainfall regression (illustrative data for a rainfed crop in one district)
Rainfall x (mm): 420, 510, 380, 600, 470, 550; yield y (t/ha): 1.18, 1.26, 1.02, 1.38, 1.08, 1.45.
$\bar{x}$ = 488.3 mm; $\bar{y}$ = 1.228 t/ha; $\Sigma(x - \bar{x})(y - \bar{y})$ = 59.88; $\Sigma(x - \bar{x})^2$ = 33,483; $\Sigma(y - \bar{y})^2$ = 0.1409.
b = 59.88 / 33,483 = 0.001788 ≈ 0.00179 t/ha per mm; a = 1.228 − 0.001788 × 488.3 = 0.355 t/ha.
r = 59.88 / √(33,483 × 0.1409) = 0.87; R^2 = 0.76 — rainfall explains about three-quarters of the year-to-year variation in yield.
Forecast: if seasonal rainfall is 500 mm, $\hat{y}$ = 0.355 + 0.00179 × 500 = 1.25 t/ha.

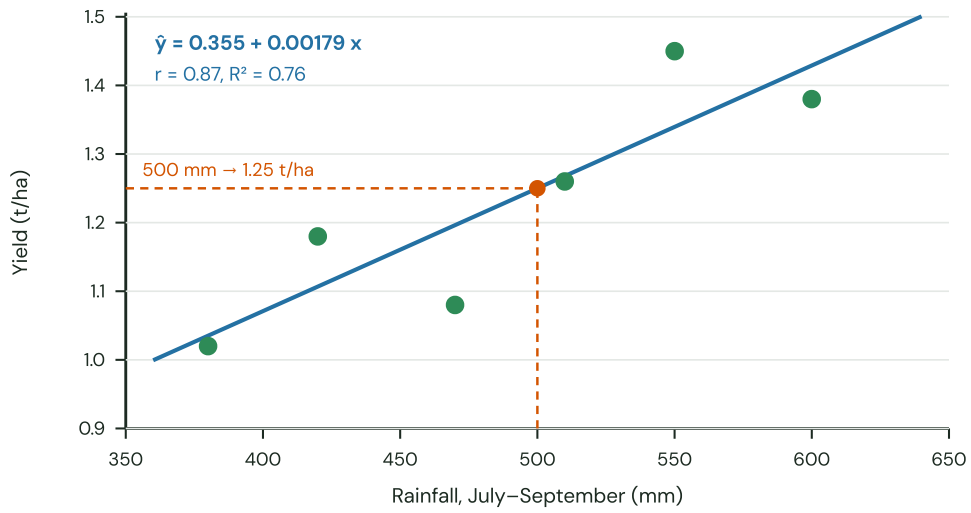


Figure 2. Yield–rainfall regression for the worked example. Original chart from illustrative data.

Regression models are reliable only within the range of the data used to fit them (here 380–600 mm), and correlation does not prove cause: other factors that vary with rainfall, such as pests or sowing dates, also affect yield.

4. Weather analysis for agriculture

Weather is the state of the atmosphere at a given time and place; **climate** is its long-term average and variability. Crop growth and farm operations depend on solar radiation, temperature, rainfall, humidity, wind and evaporation. IMD recognises four meteorological seasons over India: winter (January–February), pre-monsoon (March–May), south-west monsoon (June–September) and post-monsoon (October–December) (India Meteorological Department [IMD], n.d.).

Analysis	What it tells the farmer or engineer
Mean, standard deviation and coefficient of variation (CV)	Typical value and how much it varies from year to year — a high CV of rainfall means high risk
Moving averages and trends	Whether rainfall or temperature is changing over decades
Rainfall probability and dependable rainfall	Rainfall that can be expected in, say, 3 years out of 4 — used to plan crops and design water-harvesting structures
Return period	How often, on average, an event of a given size occurs — used to design drains, bunds and spillways
Dry and wet spells; onset of monsoon	Sowing dates, supplementary irrigation, risk of crop failure
Growing degree days	Crop development speed and expected dates of flowering and maturity
Reference evapotranspiration	Crop water requirement and irrigation scheduling (Lecture 13)

Rainfall probability

Arrange n years of rainfall in descending order and give each a rank m. The probability that rainfall will be equalled or exceeded is estimated with the Weibull formula, and the return period is its reciprocal:

$$P\,(\%) = m / (n + 1) \times 100 \qquad T\,(\text{years}) = 1 / P = (n + 1) / m$$

Worked example 3 — dependable June rainfall (illustrative, 10 years)
Ranked rainfall (mm) with P: 205 (9.1%), 171 (18.2%), 142 (27.3%), 133 (36.4%), 118 (45.5%), 110 (54.5%), 96 (63.6%), 88 (72.7%), 64 (81.8%), 52 (90.9%).
75% dependable rainfall lies between ranks 8 and 9: 88 + (75 - 72.7) / (81.8 - 72.7) × (64 - 88) ≈ 82 mm. A crop plan for June should therefore rely on about 82 mm, not the 10-year mean of 118 mm.
Return period of 171 mm or more: T = 11 / 2 = 5.5 years.

Growing degree days (heat units)

Crops develop faster in warm weather. **Growing degree days (GDD)** add up the daily heat above a crop's **base temperature** (T_base), below which development stops. A crop variety needs a roughly constant number of GDD to reach flowering or maturity, so GDD predict crop stages better than calendar days.

$$GDD = \Sigma\ [\ (Tmax + Tmin)\ /\ 2 - T_base\ ] \qquad \text{(daily values below zero are counted as zero)}$$

Worked example 4 — GDD for maize (T_base = 10 °C; illustrative temperatures)
Five days (Tmax/Tmin, °C): 32/18, 34/20, 30/16, 28/12, 33/21 → daily means 25, 27, 23, 20, 27 → GDD 15, 17, 13, 10, 17 → total = 72 degree-days.
Cold day 22/6 °C: Method 1 uses the mean, 14 °C → 4 GDD. Method 2 first raises Tmin to T_base (10 °C) → (22 + 10)/2 - 10 = 6 GDD.

The two methods give different totals whenever Tmin falls below T_base, so a report must state which method it used (McMaster & Wilhelm, 1997).

Weather forecasts and agromet advisories

IMD forecast type	Lead time / validity	Farm use
Nowcast	Less than 24 hours	Thunderstorm, hail or heavy-rain warnings; stop spraying or harvesting
Short range	1 to 3 days	Irrigation, spraying and harvest decisions
Medium range	4 to 10 days	Planning sowing, fertiliser and irrigation for the week
Extended / long range	Beyond 10 days; long-range forecasts are usually for a season	Choice of crop and variety; seasonal planning

Forecast ranges as defined in the IMD glossary (IMD, n.d.).

Under the **Gramin Krishi Mausam Sewa (GKMS)** scheme, IMD issues district- and block-level weather forecasts for the next 5 days. Its network of 130 Agromet Field Units, located at agricultural universities, ICAR institutes and other institutions, turns these forecasts into crop-specific advisories every **Tuesday and Friday**. The advisories reach farmers as bulletins and SMS, including through the mKisan portal (IMD, 2020).

5. Crop simulation models

A **crop simulation model** is a dynamic, process-based computer model that "grows" a virtual crop, usually one day at a time, from sowing to maturity. Each day it uses the weather to calculate development, growth, water use and nutrient uptake, and it carries the results forward to the next day.

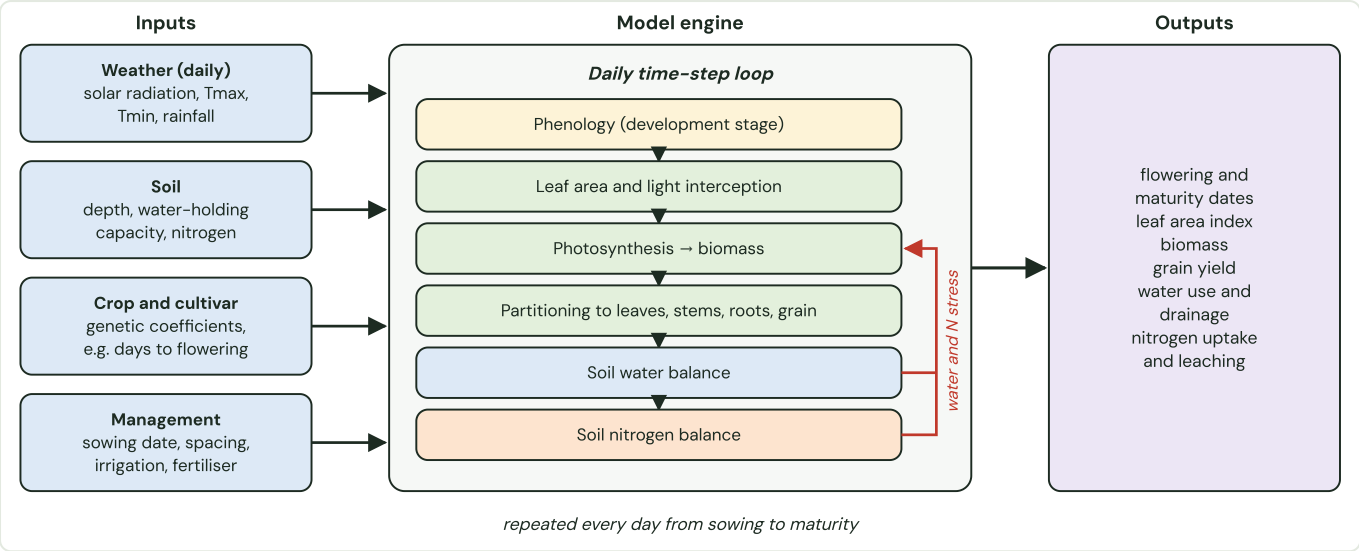


Figure 3. General structure of a crop simulation model: inputs, daily processes and outputs. Original diagram.

How a model grows biomass: light interception and radiation-use efficiency

Many crop models calculate daily growth from the light the canopy intercepts. The fraction of light intercepted rises with the **leaf area index (LAI)** — the leaf area per unit ground area — following an exponential (Beer's law) curve. Biomass produced is proportional to intercepted radiation, through the **radiation-use efficiency (RUE)**, an approach developed by Monteith (1977). Grain yield is a fraction of the total biomass, the **harvest index (HI)**.

$$f = 1 - e^{(-k \cdot LAI)} \qquad \text{Daily biomass} = RUE \times PAR \times f \qquad \text{Yield} = HI \times \text{total biomass}$$

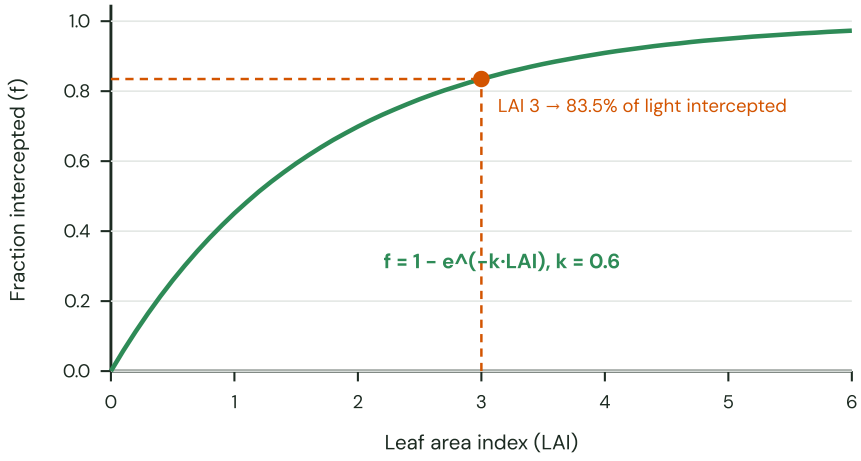


Figure 4. Fraction of light intercepted by a crop canopy as leaf area increases (k = 0.6). Original chart.

Worked example 5 — one day of crop growth (illustrative values)
Solar radiation 20 MJ/m² per day; photosynthetically active radiation (PAR) ≈ 50% = 10 MJ/m².
LAI = 3, k = 0.6 → f = 1 - e^(-1.8) = 0.835.
RUE = 3.0 g of dry matter per MJ of intercepted PAR → biomass = 3.0 × 10 × 0.835 = 25.0 g/m² per day = 250 kg/ha per day.
Over a 110-day season averaging 18 g/m² per day: 1,980 g/m² = 19.8 t/ha of biomass; with HI = 0.45, grain yield = 8.9 t/ha.
A full model would reduce growth on days of water or nitrogen stress, low temperature or low light.

Well-known crop simulation models

Model	Developed by / origin	Main features
DSSAT (Decision Support System for Agrotechnology Transfer)	International consortium of universities and institutes (Jones et al., 2003)	Software with crop models for over 45 crops (version 4.8.5), including the CERES cereal and CROPGRO legume models, plus tools for soil, weather and management data. Needs daily weather, soil profile data, crop management and cultivar coefficients (DSSAT Foundation, n.d.)
APSIM (Agricultural Production Systems sIMulator)	Australian research organisations (Holzworth et al., 2014)	A modular framework for farming systems — crops, pastures, soils, rotations and management — used for food security, climate adaptation and mitigation studies
AquaCrop	FAO (Steduto et al., 2009)	"Water-driven" growth: transpiration is calculated first and converted to biomass through water productivity; uses canopy cover instead of LAI and needs relatively few inputs. Suited to rainfed, deficit and full irrigation studies
InfoCrop	Developed in India (Aggarwal et al., 2006)	A dynamic model for tropical environments that estimates crop yields, yield losses due to pests, and environmental impacts of agro-ecosystems

Uses and limitations

Uses	Limitations
Choosing sowing dates, varieties, irrigation and fertiliser schedules	Need detailed, good-quality weather, soil and management data
Yield forecasting before harvest	Must be calibrated for local cultivars and conditions
Assessing the impact of climate change and testing adaptation options	Many models ignore or simplify pests, diseases, weeds and extreme events
Planning research: testing many "virtual" experiments before field trials	Errors in inputs and parameters carry through to outputs; results have uncertainty
Training students and extension staff in how crops respond to their environment	Require skilled users to run and interpret them

In India, the **Mahalanobis National Crop Forecast Centre (MNCFC)**, set up by the Ministry of Agriculture and Farmers Welfare in April 2012, uses satellite observations to make pre-harvest production forecasts of nine field crops under the FASAL programme. FASAL stands for Forecasting Agricultural Output using Space, Agro-meteorology and Land based Observations, and its yield forecasts are based on weather parameters or spectral indices (Indian Space Research Organisation [ISRO], n.d.).

6. Key takeaways

- A model is a simplified representation of a system; models may be empirical or mechanistic, deterministic or stochastic, static or dynamic.
- Models are built in steps and must be calibrated, then validated on independent data; MBE, RMSE and Willmott's d measure agreement.
- Statistical models — regression, ANOVA, time series and probability analysis — link yield to weather and inputs but are valid only within the range of their data.

- Weather analysis supports planning: dependable rainfall (Weibull method), return periods and growing degree days.
- IMD forecasts range from nowcasts to seasonal outlooks; GKMS advisories are issued every Tuesday and Friday from 5-day forecasts.
- Crop simulation models such as DSSAT, APSIM, AquaCrop and InfoCrop grow a virtual crop day by day from weather, soil, cultivar and management inputs.
- Light interception (f = 1 - e^(-k·LAI)), radiation-use efficiency and harvest index link weather to yield in many crop models.

7. Examination questions

Short-answer questions (2 marks)

Q1. Define a model. Distinguish empirical from mechanistic models.

A simplified representation of a system. Empirical models fit relationships to data; mechanistic models represent the underlying processes.

Q2. What is the difference between calibration and validation?

Calibration adjusts parameters to match one dataset; validation tests the model on independent data.

Q3. What does $R^2 = 0.76$ mean in a yield–rainfall regression?

Rainfall explains 76% of the variation in yield.

Q4. Write the Weibull formula for probability of exceedance.

$P = m / (n + 1) \times 100$, where m is the rank in descending order and n the number of years.

Q5. Define growing degree days.

The sum of daily mean temperature above a crop's base temperature, used to predict crop development.

Q6. State the IMD lead times for short- and medium-range forecasts.

Short range: 1–3 days; medium range: 4–10 days.

Q7. What is GKMS?

Gramin Krishi Mausam Sewa — IMD's scheme for weather-based agromet advisories, issued every Tuesday and Friday.

Q8. What is radiation-use efficiency?

Dry matter produced per unit of intercepted (photosynthetically active) radiation, e.g. in g/MJ.

Q9. Name four crop simulation models.

DSSAT, APSIM, AquaCrop and InfoCrop.

Q10. What is RMSE?

Root mean square error — the typical size of the difference between predicted and observed values.

Long-answer questions (5–10 marks) — answer outlines

Q1. Classify agricultural models and describe the steps in developing and testing a model.

Classification table (Section 1); Figure 1; calibration, validation, sensitivity analysis; MBE, RMSE, d with worked example 1.

Q2. Explain simple linear regression and fit a yield–rainfall model from given data.

Formulas; worked example 2 (b, a, r, R^2, forecast); Figure 2; limitations of regression.

Q3. Describe the methods of weather analysis used in agriculture.

Table in Section 4; Weibull probability and dependable rainfall (worked example 3); return period; GDD (worked example 4).

Q4. Explain the types of weather forecasts issued by IMD and the agromet advisory service.

Forecast table; GKMS — 5-day forecasts, 130 Agromet Field Units, Tuesday and Friday bulletins, SMS dissemination.

Q5. Describe the structure of a crop simulation model and how it calculates daily growth.

Figure 3 — inputs, processes, outputs; light interception and RUE (Figure 4, worked example 5); stresses.

Q6. Compare DSSAT, APSIM, AquaCrop and InfoCrop and discuss the uses and limitations of crop models.

Comparison table; uses (management, forecasting, climate change, research); limitations (data, calibration, uncertainty); MNCFC and FASAL.

8. Glossary

Term	Meaning
Calibration	Adjusting model parameters so that outputs match observations
Climate	Long-term average and variability of weather
Crop simulation model	A dynamic, process-based model that simulates crop growth, usually day by day
Cultivar coefficients	Parameters describing a variety, such as its development rate
Dependable rainfall	Rainfall expected to be equalled or exceeded with a stated probability, e.g. 75%
Deterministic model	Same inputs always give the same output
Dynamic model	A model whose state changes over time
Empirical model	A model fitted to data without representing processes
GDD	Growing degree days — accumulated heat above a base temperature
Harvest index	Fraction of total biomass that is harvested as economic yield
LAI	Leaf area index — leaf area per unit ground area
Mechanistic model	A model based on the underlying processes
PAR	Photosynthetically active radiation
Regression	A statistical method that predicts one variable from others
Return period	Average interval between events of a given size
RMSE	Root mean square error
RUE	Radiation-use efficiency — biomass per unit of intercepted radiation
Sensitivity analysis	Testing how much outputs change when an input or parameter changes
Stochastic model	A model that includes random variation
Validation	Testing a model against independent observations

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All diagrams and charts are original. Explanations are written in original language. Data in the worked examples are illustrative, and every result shown was recalculated.

Practice MCQs

Q1. A model that fits a relationship to observed data without representing the underlying processes is

(a) mechanistic (b) empirical
(c) stochastic (d) dynamic

Q2. A model whose state variables change day by day is

(a) static (b) dynamic
(c) descriptive (d) empirical

Q3. Testing a calibrated model against independent data is called

(a) calibration (b) sensitivity analysis
(c) validation (d) normalisation

Q4. In the regression $\hat{y} = a + bx$, b is the

(a) intercept (b) slope
(c) correlation coefficient (d) residual

Q5. If $R^2 = 0.76$ for yield regressed on rainfall, rainfall explains

(a) 0.76% of the variation in yield (b) 24% of the variation in yield
(c) 76% of the variation in yield (d) all of the variation in yield

Q6. The Weibull formula for probability of exceedance is

(a) m / n
(b) m / (n + 1)
(c) (n + 1) / m
(d) n / m

Q7. With a base temperature of 10 °C, a day with Tmax 30 °C and Tmin 16 °C contributes

(a) 13 GDD
(b) 23 GDD
(c) 20 GDD
(d) 6 GDD

Q8. IMD medium-range forecasts have a lead time of

(a) less than 24 hours
(b) 1 to 3 days
(c) 4 to 10 days
(d) a whole season

Q9. Under GKMS, agromet advisories are issued

(a) daily
(b) every Tuesday and Friday
(c) once a month
(d) once a season

Q10. Which crop model is "water-driven", converting transpiration to biomass?

(a) DSSAT
(b) APSIM
(c) AquaCrop
(d) InfoCrop

Q11. Which crop model was designed for tropical environments and also estimates yield losses due to pests?

(a) InfoCrop
(b) AquaCrop
(c) CERES
(d) APSIM

Q12. In crop models, the fraction of light intercepted by the canopy is calculated mainly from

(a) soil pH
(b) leaf area index
(c) rainfall
(d) harvest index

Answers

Q1 — (b) empirical
Q2 — (b) dynamic
Q3 — (c) validation
Q4 — (b) slope
Q5 — (c) 76% of the variation in yield
Q6 — (b) m / (n + 1)
Q7 — (a) 13 GDD
Q8 — (c) 4 to 10 days
Q9 — (b) every Tuesday and Friday
Q10 — (c) AquaCrop
Q11 — (a) InfoCrop
Q12 — (b) leaf area index